DESCRIPTION OF COURSEWORK

Course Code	EEE301
Course Name	Signals and Systems
Lecturer	Siti Balqis Samdin
Academic Session	2023/04
Assessment Title	Laboratory Assignment

A. Introduction

This laboratory practical applies some of the theoretical principles learned throughout the lecture. Thus, the task given is in an open-ended format. Students are given the freedom to design the experiments instead of merely following a set of guidelines or manual.

B. Course Learning Outcomes (CLO) covered

At the end of this assessment, students are able to:

CLO 4 Apply modern simulation tools in analysing and designing practical circuits (PLO5).

C. University Policy on Academic Misconduct

1. Academic misconduct is a serious offense in Xiamen University Malaysia. It can be defined as any of the following:
   i. **Plagiarism** is submitting or presenting someone else's work, words, ideas, data or information as your own intentionally or unintentionally. This includes incorporating published and unpublished material, whether in manuscript, printed or electronic form into your work without acknowledging the source (the person and the work).
   ii. **Collusion** is two or more people collaborating on a piece of work (in part or whole) which is intended to be wholly individual and passed it off as own individual work.
   iii. **Cheating** is an act of dishonesty or fraud in order to gain an unfair advantage in an assessment. This includes using or attempting to use, or assisting another to use materials

that are prohibited or inappropriate, commissioning work from a third party, falsifying data, or breaching any examination rules.

2. All the assessment submitted must be the outcome of the student. Any form of academic misconduct is a serious offense which will be penalised by being given a zero mark for the entire assessment in question or part of the assessment in question. If there is more than one guilty party as in the case of collusion, both you and your collusion partner(s) will be subjected to the same penalty.

D. Instruction to Students

1. This task is a group laboratory task. You are required to form a GROUP of TWO (2) to THREE (3) students to complete the task. Choose your own partner.
2. The report MUST be in a typewriting format. Include the cover page with 'own work declaration' statements.
3. Complete all tasks accordingly.
4. A group leader is required to submit the completed task onto MOODLE site. There will be two types of files to be submitted in MOODLE;
   i. Report
   ii. MATLAB codes/Octave codes, etc.
   (*All files submitted in .zip folder*)
5. The due date is on June 14th, 2023 (Wednesday) by 11:59 pm.

E. Evaluation Breakdown

No.	Component Title	Percentage (%)
1.	Report	60
2.	Simulation code/materials	40
	TOTAL	**100**

F. Task(s)

Title: Analysing continuous linear time-invariant (LTI) system.

Objectives:

1. To determine the frequency response of respective resistant-capacitor (RC) circuits.
2. To analyse the output response of the LTI system for various type of inputs (such as, square wave, triangular wave, and causal sinusoid function)

Introduction:

The main interest of a linear time-invariant system is calculating the output response on an applied input to the system. The relationship between output and input as a function of signal frequency ω is known as frequency response. Generally, the frequency response of an LTI system is expressed in magnitude and phase as follows:

$$H(j\omega) = |H(j\omega)|e^{j\angle H(j\omega)}.$$

One of the most straightforward LTI systems is resistor-capacitor circuits. This circuit is usually applied as a subsystem of a more extensive system that functions as a filter.

Experimental Principles:

The series resistant-capacitor (RC) circuits are as follows:

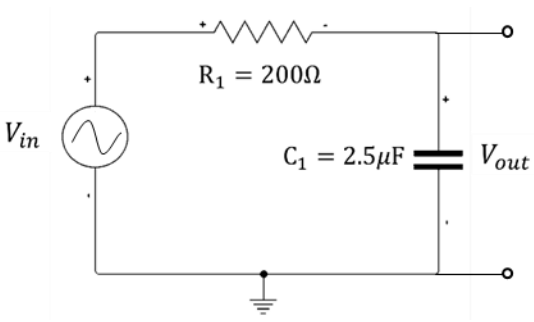


Figure 1

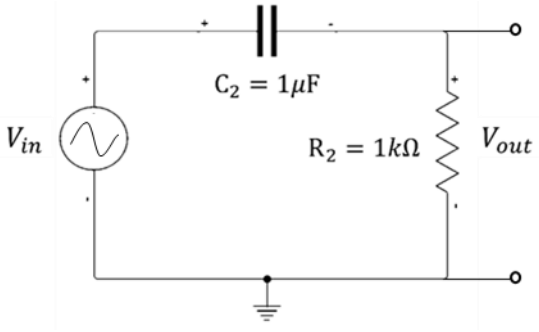


Figure 2

Components:

1. Capacitors: $C_1 = 2.5\mu F$; $C_2 = 1\mu F$.
2. Resistors: $R_1 = 200\Omega$; $R_2 = 1k\Omega$.

Figure 1 and Figure 2 shows the experimental circuits that needs to tested, where V_{in} is the input signals and V_{out} is the output. Based on the given circuits, design the experiments to meet the given objectives. Then, conduct the appropriate analysis and discuss the finding. Show the results appropriately in the form of tables, graphs and etc.

A few software can be used to simulate the experiment, such as MATLAB, OCTAVE or any other relevant software.

Warning: This an open-ended laboratory assignment with many solutions. A similar design/code/script will be considered plagiarism. Plagiarism of any kind will not be tolerated.

Report guidelines:

The formal report must contain the following standard content:

Title, Objective, Introduction, Methodology, Results, Discussion, Conclusion and References.

Your report shall not exceed 15 pages (exclude appendices, front pages) with line-spacing 1.5, Font type Times New Roman, Font size 12pt, and paragraph alignment Justified.

APPENDIX

MARKING RUBRICS

Component Title	Report and Coding Simulation				Percentage (%)	100	
Criteria	Score and Descriptors					Weight (%)	Marks
	Excellent (5)	Good (4)	Average (3)	Need Improvement (2)	Poor (1)		
Introduction	Excellent content of introduction written that is related to the title and objectives.	Acceptable content of introduction written that is related to the title and objectives.	The introduction written is somewhat related to the title and objectives.	The introduction is not related to the title and objectives.	Unrelated introduction.	**10**	
Methods	Excellent methods proposed to achieve the objectives, and well stated and described methods.	Correct methods presented to achieve the objectives and satisfactorily stated and described methods.	Somewhat correct methods presented to achieve the objectives, and also incomplete statement and description.	Incorrect methods proposed to achieve the objectives, and there is a description of the methods.	Irrelevant methods used/proposed .	**15**	
Results	Excellently presented experiment data and results with the aid of tables, figures, plots etc.	Satisfactorily presented data results and with the aid of tables, figures, plots etc.	Presented data results are somewhat reasonable with the aid of tables, figures, plots etc.	There are data results tables, figures, plots etc., with unsuitable title and description.	Unrelated results and data presented	**15**	
Discussions	Excellent analysis of data and results obtained, and well relate/compare the finding with the theoretical principles.	Satisfactorily analysis of data and results obtained, and able to relate/compare the finding with the theoretical principles.	Somewhat good analysis of data and results obtained, and with discussions.	Analysis of results and data is inappropriate with unsuitable discussion.	Unrelated analysis and discussion.	**15**	
Conclusion and References	Excellent conclusion with intelligently summarised the findings by relating the objectives, methods, results etc. Excellent references that are related to the task with proper citation.	Satisfactory conclusion statements with a good summary of the findings by relating the objectives, methods, results etc. Satisfactory references that related to the task with some inappropriate citation.	An average conclusion statement with a summary of the findings can somewhat relate to the objectives, method etc. Some of the references are unrelated and incorrect citation.	Inappropriate conclusion and summary. References are unrelated and no citation.	Unrelated conclusion. Irrelevant references.	**5**	
Coding related materials	Excellent written codes, diagrams, algorithms etc., well functioned and easy to understand	Good written codes, diagrams, algorithms etc. and functioned.	Messy written codes, diagrams, algorithms etc. and functioned.	Messy written codes, diagrams, algorithms etc. and not functioned.	No simulation materials attached.	**40**	
					TOTAL	**100**	

Note to students: Please print out and attach this appendix with the submission of coursework